STAFF REVIEW · H446 107 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Any two: process management, memory management, files, devices, security or interface.

2. O(n).

3. B, because a stack is LIFO.

4. It discards half the search space using an ordering comparison.

I Do — source worked steps

1. Model answer — Page number = 2 and offset = 452. Frame 7 base is 7 × 1,024 = 7,168; physical address = 7,620, assuming that page is mapped to frame 7. — Check: link each claim to the scenario and use precise OCR terminology.

We Do — guided CHECK

1. What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.
Work through the method
Use the middle stage of the visual to follow the process and state each intermediate result.
Shared worked answer
4,100 ÷ 1,024 gives page 4 and offset 4. Frame 1 starts at 1,024 so physical address = 1,028.

You Do — source mark schemes

1. Paging uses fixed-size blocks [2], segmentation variable-size logical blocks [2], and both support memory mapping/protection/management [2].

2. Page = 2 [2]; offset = 1 [2], as 2×4,096 = 8,192.

3. Too few resident pages [2]; page faults increase [2]; frequent disk/SSD transfers are slower than RAM [2]; reduced useful CPU execution and mitigation via more RAM/reduced working set [2].

Exit Ticket — source answer

1. False. Page faults can require slow secondary-storage access.